AMT Worksheet 10
Dr. Osborne 2023

The purpose of this problem set is to give you some practice with the definition of the entropy as well as using the Maxwell relations and other differential manipulations to relate thermal derivatives to one another. Throughout this assignment, please use the symbol $B = -V\left(\partial p/\partial V\right)_T$ for the isothermal bulk modulus, $\alpha = \left(\partial V/\partial T\right)_p \big/ V$ for the isobaric thermal expansion coefficient, and $C_p = T\left(\partial S/\partial T\right)_p$ for the heat capacity at constant pressure.

1. A system of $N = 3\times10^{24}$ atoms can exist in three quantum states associated with energies of 0, ε, and 4ε. The total energy of the system is $E = N\varepsilon/2$. Please use the expression $S = k_B \ln\Omega \approx k_B\left[-\sum_{k=1}^{3} n_k \ln\left(\frac{n_k}{N}\right) + \frac{1}{2}\ln\left(\frac{N}{4\pi^2 n_1 n_2 n_3}\right)\right]$ for the entropy throughout, where the subscripts refer to the energy states. Give all answers to these questions as numerical values complete with units.
   (a) Suppose there are no atoms in the highest energy state. How many atoms are in each of the other two states in this case?

   (b) Explain why the expression given above for the entropy is flawed in the case associated with part (a). Then, modify part (a) slightly by supposing instead that there are 20 atoms in the highest energy state. Use the formula above to determine the entropy in this case.

   (c) Determine the entropy of the system in the case of 100 atoms in the highest energy state.

   (d) Compare your results from parts (b) and (c) and explain why both of these situations can be considered *minor* variations from part (a).

   (e) Determine the ratio of the second contribution to the entropy (the one with π) to that of the sum in the scenario given in part (b). Explain why the result from part (c) will be similar and what your result means about the *importance* of the second contribution to our approximation of the entropy.

   (f) Suppose there are $N/100$ atoms in the highest energy state. How many atoms are in each of the other two states in this case? Determine the entropy of the system in this scenario. Explain the difference between this result and those from parts (b) and (c). Does this represent a *minor* variation from part (a)?

(g) Suppose there are $N/25$ atoms in the highest energy state. How many atoms are in each of the other two states in this case? Determine the entropy of the system in this scenario. Explain the difference between this result and that from part (f). Did the entropy increase or decrease?

(h) Suppose there are $N/10$ atoms in the highest energy state. How many atoms are in each of the other two states in this case? Determine the entropy of the system in this scenario. Explain the difference between this result and that from part (g). Did the entropy increase or decrease? What is the difference between this change and those seen previously? What does this indicate about the number of atoms in the highest energy state associated with the *maximum* entropy for this total energy? Use the Boltzmann factor to confirm your statement by calculating the fraction of the total number of atoms in each state at maximum entropy.

(i) Determine the *absolute* maximum entropy this system can have, without regard for the total energy. What is the total energy of this 'maximum entropy' system? Give this last answer as a decimal multiple of $N\varepsilon$.

2. Ocean water has a bulk modulus of about 2.34 GPa, or 2.34×10^{9} Pa, at ordinary pressures. This is approximately 23,100 atm. The pressure increases as we descend into a body of water (or any liquid) as $\Delta p = \rho g h$, where h is the depth of the water. Assuming the bulk modulus is constant (it's not actually… it increases with pressure), calculate the fractional change in volume associated with water at the bottom of the Pacific Ocean (4 km down) in comparison to water at the surface. Use $1025\ \mathrm{kg/m^3}$ for the density.

3. The thermal expansion coefficient of steel is about $13\times10^{-6}/\mathrm{K}$. Using this, determine the fractional change in volume of a steel beam as it heats up from 0 degrees Celsius to 40 degrees Celsius. Explain why this might be relevant to structural engineers designing railroads.

4. Determine the value of B, α, and C_p for n moles of an ideal gas with ℓ degrees of freedom held at pressure p and temperature T.

5. One of the most difficult derivatives to express in terms of our three chosen quantities, and a sort of 'key' to getting the rest of the derivatives, is the heat capacity at constant volume, $C_V = T\left(\partial S/\partial T\right)_V$. Determine the value of this derivative in terms of any or all of *p*, *T*, *V*, *B*, α, C_p by thinking of *S* as a function of *p* and *T*, dividing its differential by *dT*, and taking the volume as constant.

6. Determine the following derivatives in terms of any or all of *p*, *T*, *V*, *B*, α, C_p and state whether you expect them to be positive or negative for an ordinary material. Note: for these problems, it is easiest to think of *S* as representing heat, i.e. you should think of changes in *S* as representing a flow of heat; constant entropy processes are those taking place with no heat flow into or out of the system.
   (a) $\left(\partial V/\partial S\right)_p$.

   (b) $\left(\partial T/\partial V\right)_S$.

   (c) $\left(\partial p/\partial S\right)_V$.

   (d) $\left(\partial T/\partial p\right)_S$